# Multimodal AI Product Evaluation — 101

### A Beginner's Guide to Testing AI Products That See, Hear, Read, and Speak

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## How to Use This Guide

This guide is designed to be read in order, at least the first time through. Each part builds on the previous one. To help you pace yourself, each section is marked with a difficulty indicator:

🟢 **No background needed** — readable by anyone

🟡 **Light background** — helpful to have read Parts 1-5 first

🔴 **Some technical familiarity** — terms from earlier sections assumed

Every concept is introduced in plain language before any jargon appears. If you hit a term you don't recognize, the Quick Reference Glossary at the end defines everything.

**If you don't have a product yet:** Most of this guide applies to you. The "First Steps" section at the end has a specific path for pre-launch teams.

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## Start Here: What Problem Are We Solving? 🟢

Imagine you hired a research assistant. You hand them a stack of documents, some audio recordings, and a few charts, and you ask them to prepare a 10-minute summary podcast for you.

You come back and the podcast sounds *brilliant* — confident, well-paced, professional. But then you check one of the claims against your documents. It's wrong. The assistant guessed at a chart they couldn't read clearly, got the number wrong, and stated it as fact.

**That is the core problem with multimodal AI products.**

They are impressive-sounding and impressive-looking — often before they are accurate. The gap between "sounds good" and "is good" is exactly what multimodal AI evaluation is designed to close.

The good news: the AI industry has been building tools, frameworks, and hard-won practices for exactly this problem. This guide distills the most important ideas into a practical foundation —

drawing on platforms like Galileo, Arize AI, Braintrust, Comet Opik, Weights & Biases Weave, Patronus AI, LangSmith, and research from Stanford, DeepMind, and Meta.

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# Part 1: The Basics — What is a "Multimodal" AI Product? 🟢

## Single-mode vs. Multi-mode

A **text-only AI** deals with one type of content: words. A chatbot reads your question and writes an answer. Simple input, simple output.

A **multimodal AI** deals with multiple types of content — called *modalities* — at the same time:

| Modality | Examples |
|---|---|
| Text | Documents, articles, transcripts |
| Images / Visual | Charts, diagrams, photographs, slides |
| Audio | Podcasts, voice recordings, meeting recordings |
| Video | Lectures, explainer videos, screen recordings |
| Documents | PDFs, Word files, scanned pages |
| Structured data | Tables, spreadsheets, databases |

A multimodal AI product might take in a PDF report, a YouTube video, and a set of slides, then produce an audio summary, a visual diagram, and a written brief — all from the same request.

The more modalities involved, the more powerful the product — and the more places things can go wrong.

## Where the Industry Currently Sits

When evaluation platforms like Galileo, Arize, and Braintrust launched in 2022-2023, they focused primarily on text-only AI. By 2025-2026, all had extended to handle multimodal content — document parsing quality, audio transcription accuracy, visual grounding, and cross-modal consistency. This guide reflects that matured thinking.

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# Part 2: Why Evaluating Multimodal AI is Harder 🟢

## The "It Sounds Great" Trap

With text-only AI, you read the output and check if it makes sense. Quality and accuracy are visible in the same medium.

With multimodal AI, **presentation quality and content accuracy are separated.** An audio output can have excellent pronunciation, natural pacing, and a warm tone — while simultaneously getting facts wrong because it couldn't read a chart in the source document.

Human listeners instinctively trust polished audio. Research consistently finds that people rate audio quality higher than content accuracy — until they're forced to evaluate them separately. This is the trap.

> **Galileo Research, 2025:** In their analysis of multimodal AI user testing, Galileo found that polished presentation was the single strongest predictor of users *not* flagging errors — people trusted what sounded professional.

## The Evaluation Paradox

Galileo researchers describe what practitioners call the **evaluation paradox**: generating AI content is often easier than evaluating it. A language model produces a convincing 10-minute audio summary in seconds — but verifying whether every claim is grounded in source documents requires careful, systematic work.

This is why evaluation cannot be an afterthought. Arize AI's documentation makes the point explicitly: teams that instrument evaluation from the beginning catch failures orders of magnitude cheaper than teams that bolt it on after launch.

## The Seams Problem

Think of a multimodal AI system like a factory assembly line. Each station does one job: one reads your PDF, another extracts figures, another finds relevant content, another writes a script, another converts that script to audio.

The finished product is the result of all stations working together. When something goes wrong, the defect may have been introduced at **any station** — and by the time it shows up in the final output, it's been through several more stations that made it look polished.

These hand-off points are called **junctions**. Arize AI calls them **spans** in their tracing documentation. Most failures in multimodal AI products happen at the junctions, where one type of content is converted to another.

**This is why evaluating only the final output tells you very little.** The industry now calls the alternative **trace-based evaluation** — evaluating each stage and each hand-off independently.

### The Hidden Failure (A Concrete Pattern)

1. A PDF parser fails to extract the caption under Figure 3 ("Quarterly revenue, FY2024")
2. The AI needs to discuss Figure 3 but has no caption
3. Without context, it guesses — and says "FY2023"
4. The text-to-speech engine converts this to audio confidently
5. The output sounds perfect. The fact is wrong.

The failure originated at step 1. It only became visible in step 4. If you only check the audio, you'd try to fix the wrong thing.

Galileo built a specific "data quality" evaluation tier for exactly this reason — checking source material integrity before any model-level evaluation runs.

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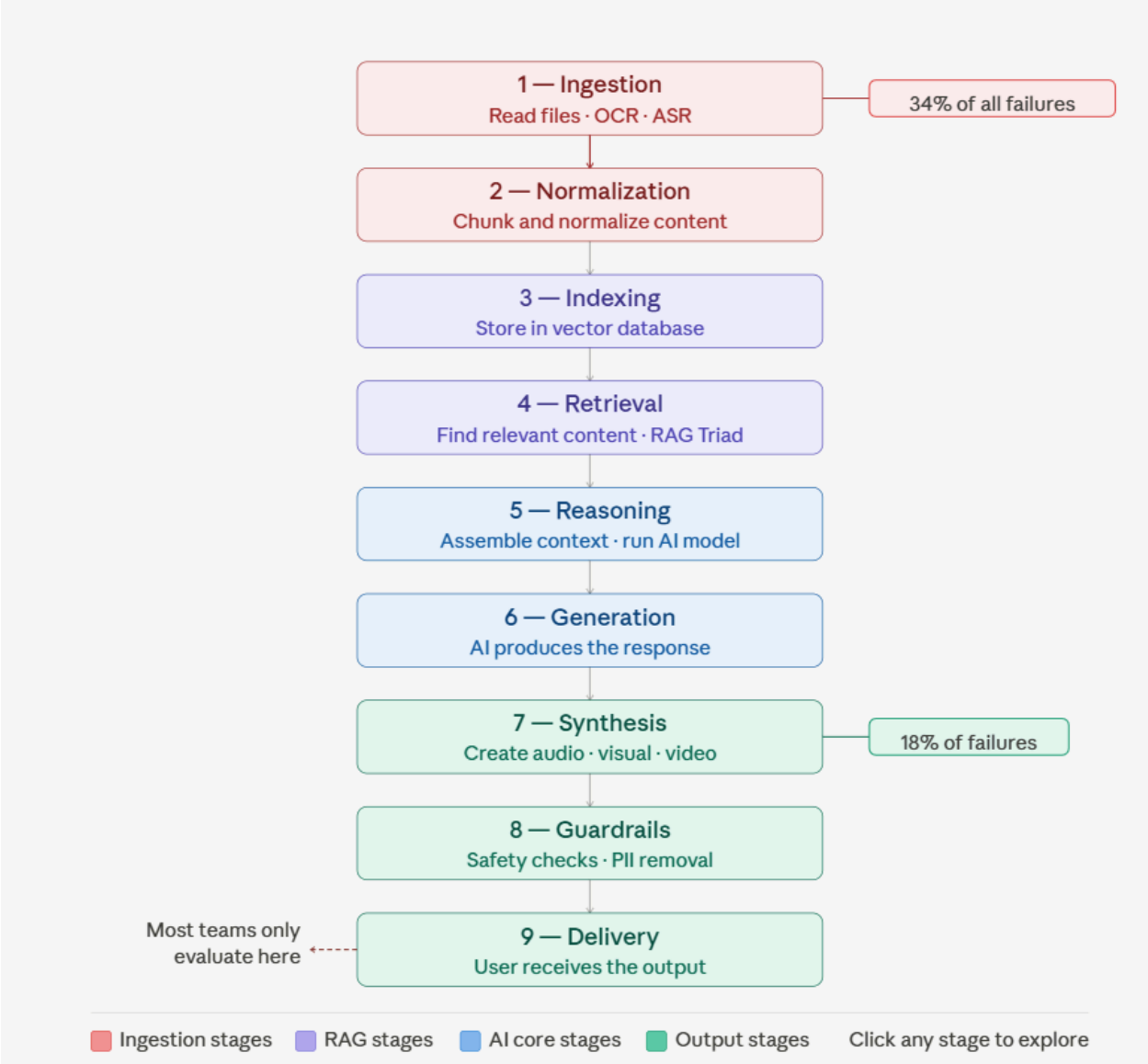
## Part 3: The Pipeline — Understanding the Full Journey 🟢

Every multimodal AI product has a **pipeline**: a sequence of steps transforming inputs into outputs. Understanding this pipeline is the foundation of good evaluation.

**Figure 1: The 9-Stage Pipeline**




STAGE 1 — INGESTION
Reading uploaded files: PDFs, audio, video, slides, images
↓ [ ~34% of all failures originate here ]
STAGE 2 — NORMALIZATION
Converting to consistent format; breaking into chunks
↓
STAGE 3 — INDEXING
Storing content for efficient search (vector database)
↓
STAGE 4 — RETRIEVAL
Finding the relevant content for a given query
↓ [ RAG Triad checks happen here ]

STAGE 5 — REASONING
  Assembling context and running the AI model
  ↓
STAGE 6 — GENERATION
  AI produces its response: text, script, diagram plan
  ↓
STAGE 7 — SYNTHESIS
  Converting to final form: audio, visual, video
  ↓ [ ~18% of failures originate here ]
STAGE 8 — GUARDRAILS
  Safety checks: private info removal, content filtering
  ↓
STAGE 9 — DELIVERY
  Final output reaches the user

**Most of the evaluations are only at Stage 9.** This is like a restaurant checking food quality only after the customer has eaten. The highest-leverage approach is to evaluate each stage.

> **Beginner's Rule of Thumb:** When something goes wrong, always suspect Stage 1 (Ingestion) first. Approximately one-third of all multimodal failures originate there — before the AI model has been involved at all.

## The RAG Stages — A Closer Look 🟡

Stages 3 and 4 have their own well-developed evaluation discipline. Systems that retrieve content before generating a response are called **RAG (Retrieval-Augmented Generation)** systems.

Three organizations — TruEra (now Scale AI), Galileo, and the open-source Ragas project — independently converged on the **RAG Triad**: three things to always check:

| RAG Triad Component | The Question It Answers |
|---|---|
| **Context Relevance** | Did retrieval actually find the right content? |
| **Groundedness** | Is the AI's response based on what was retrieved? |
| **Answer Relevance** | Does the response answer what the user actually asked? |

A system can fail on any of these independently. Arize AI's Phoenix documentation adds a fourth dimension for multimodal: **Modality Completeness** — did retrieval surface content from all relevant modality types, or did it systematically ignore visual content?

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# Part 4: The Five Ways Things Go Wrong — The Five Gulfs 🟢

A "gulf" is a gap where failures originate — a misunderstanding between parts of your system. There are five.

**Figure 2: The Five Gulfs**

| Gulf name | The gap | How to bridge it |
|---|---|---|
| 1. Comprehension<br>Developer ↔ Data | Developer assumes clean inputs.<br>Real users upload scanned, noisy, handwritten, or accented content. | Study 50 real user sessions before writing any test cases. |
| 2. Specification<br>Developer ↔ Pipeline | "Summarize this" ≠ "faithfully summarize using only claims in the document, without inventing connections." | Write scoring rubrics before writing any code (Braintrust). |
| 3. Generalization<br>Data ↔ Pipeline | Works perfectly on test cases.<br>Fails when users upload legal contracts, medical reports, or new document types. | Cover the full range of real inputs.<br>Refresh test sets every quarter. |
| 4. Modality<br>Modalities ↔ each other | Audio says "revenue up 15%."<br>Diagram shows revenue declining.<br>Same system, contradicting outputs. | Build checks that compare outputs across modalities, not just within. |
| 5. Ingestion<br>new · highest leverage<br>Input Media ↔ AI | OCR misread "12%" as "72%."<br>Every downstream stage runs perfectly — on wrong data. | Measure text extraction quality before the AI sees anything.<br>Highest ROI evaluation step. |

st in text-only AI too. Gulfs 4 (Modality) and 5 (Ingestion) are unique to multimodal systems and require separate instrumentation.

Comprehension · Specification · Generalization · Modality · Ingestion (new, highest leverage)

## Gulf 1: Comprehension 🟢

**The gap between what developers think users do and what users actually do.**

Your team tests with clean PDFs from academic journals. Real users upload scanned textbooks with coffee stains, photos of handwritten notes, and 90-minute YouTube lectures in accented English.

Braintrust describes this as "distribution shift between your golden dataset and real traffic." Their recommended practice: pull 50 real user sessions and study them *before* writing a single test case.

**How to bridge it:** Test with what users actually give you, not what you have available.

## Gulf 2: Specification 🟢

**The gap between what you meant to build and what you actually built.**

You tell the AI: "summarize these documents." You really meant: "create a faithful summary using only claims present in the documents, without inventing connections, while flagging uncertainty." That difference is enormous.

LangSmith calls this the "vague rubric" problem. Without a precise definition of "good," different team members evaluate the same output differently — and automated evaluators have no reliable ground truth.

**How to bridge it:** Write down exactly what "good" means before building anything. Braintrust calls this writing a **scoring rubric** before writing code.

## Gulf 3: Generalization 🟢

**The gap between what works on your test cases and what works in the real world.**

Your product handles academic papers perfectly. A user uploads a legal contract — it falls apart. Your test cases never included legal documents.

Arize AI calls this "dataset staleness" — your evaluation set was built from what users uploaded six months ago, not today.

**How to bridge it:** Build test sets covering the full range of real-world inputs. Refresh them quarterly.

## Gulf 4: Modality 🟡

**The gap between what different modalities say.**

The audio says: "Revenue increased by 15%." The video shows revenue decreasing. The same AI system produced both — and they contradict each other.

Galileo's multimodal evaluation framework calls out "modality coherence" as a required evaluation dimension — not just whether each output is individually correct, but whether they tell a consistent story.

**How to bridge it:** Specific checks that compare outputs *across* modalities, not just within each one.

## Gulf 5: Ingestion 🟡

**The gap between what the document actually says and what the AI thinks it says.**

A low-resolution scan causes OCR to misread "12%" as "72%." Every downstream stage works perfectly — but on wrong data. The AI faithfully summarizes information that was never in the document.

**How to bridge it:** Measure the quality of text extraction before the AI gets involved. This is the most commonly overlooked step and the one with the highest ROI.

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# Part 5: Four Types of "Good" — The Contracts 🟢

Multimodal AI products have **multiple, separate quality contracts** — not a single accuracy score. Beginners often ask "is the output good?" — but "good" means different things in different dimensions. One score hides failures that matter.

## The Industry's Evolving Language

Different platforms use different terms for the same underlying concepts:

| Concept | Galileo calls it | Ragas calls it | Arize Phoenix calls it |
|---|---|---|---|
| Does output match sources? | Context Adherence | Faithfulness | Groundedness |
| Does output match reality? | Factual Accuracy | — | Factuality |
| Did retrieval find the right content? | Context Relevance | Context Precision/Recall | Context Relevance |
| Does the response answer the question? | Completeness | Answer Relevance | Answer Relevance |

## Contract 1: Faithfulness 🟢

**Does the output only say things that are actually in the source material?**

If the user uploads a report, the AI should only make claims supported by that report. If it says something not in the report — even if true in the world — that is a faithfulness failure.

*Example:* The report doesn't mention North America. The AI says "North America was the strongest region." May be true in reality, but wasn't in the source. **Faithfulness: fail.**

Galileo measures this with **Context Adherence**. Ragas measures it with **Faithfulness**. Both decompose the output into individual claims and check each against the source.

## Contract 2: Factuality 🟡

**Does the output say things that are true about the world?**

This is *different* from faithfulness. Faithfulness = grounded in the source. Factuality = accurate against reality.

*Example:* The source document has a typo — it says the product launched in 2014 when it actually launched in 2024. A faithfully grounded AI repeats the wrong date. **Faithfulness: pass. Factuality: fail.**

Patronus AI's **Lynx** model (2024) was specifically built to evaluate factuality — cross-referencing claims against world knowledge rather than just the provided context. This is harder than faithfulness and requires a more capable judge model.

**These two contracts require separate tests.** A single accuracy score collapses them and masks the difference — critical in healthcare or finance where factual errors cause real harm.

## Contract 3: Cross-Modal Consistency 🟡

**Do the different modalities agree with each other?**

If the AI produces both an audio summary and a visual diagram, they should tell the same story. If they contradict each other, that's a cross-modal consistency failure.

This contract has no equivalent in text-only evaluation. Arize AI's Phoenix evaluates this at the span level — comparing outputs of the synthesis stage across modality branches to detect contradictions.

**Important:** Two outputs can be cross-modally *consistent* (they agree with each other) while *both* being wrong. The worked example in Part 7 shows exactly this failure.

## Contract 4: Output Quality 🟢

**Is the output well-produced in its medium?**

- **Audio:** Natural sound, correct pronunciation, good pacing, no glitches
- **Visual:** Readable diagram, correct node placement, complete labels
- **Video:** Visuals aligned with narration, captions in sync

This is evaluated per modality. Comet Opik tracks these as separate metric families in their dashboard, preventing the common mistake of averaging audio and visual quality into a single score.

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# Part 6: The Build Path and Run Loop — Two Modes of Working 🟢

Every AI product goes through two phases. Understanding the difference determines what kind of evaluation you should be doing right now.

## The Build Path: Building the Feature

Step 1 → Define the goal and what "good" means (the four contracts)
Step 2 → Build a first version (prototype)
Step 3 → Define quality targets per contract
Step 4 → Create test cases and scenarios
Step 5 → Build the full version and validate it
Step 6 → Launch to users

This happens once per feature. Braintrust calls this **offline evaluation** — running against a fixed dataset before any user sees the product.

## The Run Loop: Keeping It Working After Launch

Users use the product
↓
Collect traces of what's going wrong
↓
Run evaluations on sampled traces
↓
Monitor trends, get alerts
↓
Diagnose why evaluations failed
↓
Release an improved version
↓
(repeat forever)

Arize AI calls this **online evaluation**. Galileo calls it **production monitoring**. The terminology differs; the discipline is the same.

Most beginner mistakes happen because teams only think about the Build Path. The Run Loop is where the product either compounds in quality — or slowly degrades as the world changes around it.

### Why Products Degrade Without the Run Loop

Weights&Biases Weave's documentation identifies four drift patterns that affect AI products post-launch — all happen without any code change on your part:

**Input drift:** Users start uploading different content types than they did at launch. Your evaluations stop reflecting real traffic.

**Model drift:** Your AI provider silently updates the underlying model. Behavior changes in ways you didn't anticipate.

**Evaluator drift:** The AI model powering your automated evaluators gets updated. Your evaluators start scoring differently without warning.

**Consumption drift:** Same output quality, dropping user engagement — usually signals novelty has worn off and the product isn't delivering sustained value.

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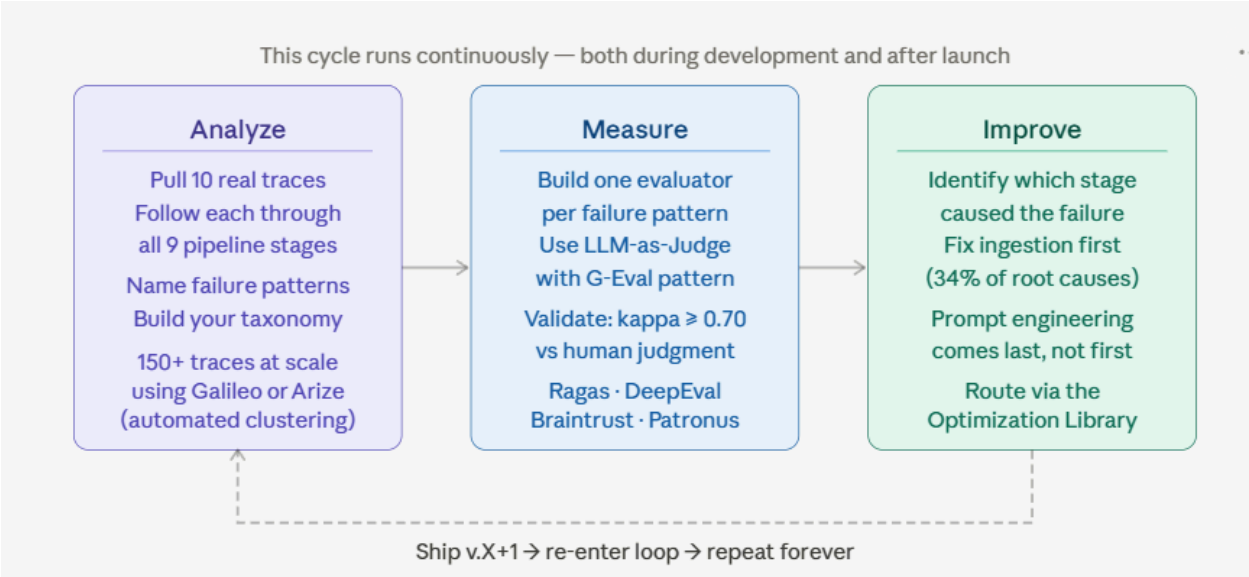
## Part 7: How to Actually Evaluate — The Three-Phase Cycle 🟢

The practical work follows three phases: Analyze, Measure, Improve. This cycle repeats continuously.

**Figure 3: The AMI Cycle**




### Phase 1: Analyze — Look at What's Actually Failing

**Manual trace review** is the foundational technique. Pull 10 real examples and examine each carefully:

- What did the user upload?
- What did ingestion produce from those uploads?
- What did retrieval actually find?
- What was the final output?
- Where did the failure first appear?

Braintrust recommends 10 deep traces over 100 shallow ones. Their rationale: 10 careful traces generate more actionable insights.

**Building a failure taxonomy**

After reviewing traces, name each recurring pattern. Your collection of named failure patterns is your **taxonomy** — the team's shared language for discussing failures. Galileo calls this a "failure mode library" — institutional memory of how your specific system breaks.

At scale, Galileo's platform clusters similar failures automatically, so you can review 50 traces in the time it took to review 5. Arize Phoenix does this visually — failures cluster in embedding space, making problematic input types immediately visible.

## Phase 2: Measure — Build Automated Checks 🟡

**What evaluators are called across different platforms:**

| Platform | Term | What it does |
|---|---|---|
| Braintrust | Scorer | A function that grades an output |
| Galileo | Metric | An automated quality measure |
| Arize Phoenix | Evaluator | A judge that labels outputs |
| Comet Opik | Metric | A scoring function per dimension |
| W&B Weave | Scorer | A function returning a score |
| LangSmith | Evaluator | An automated judge |

Despite different names, they all do the same thing: take an AI output (plus usually the input and source) and return a score or label.

**LLM-as-Judge: The Modern Standard**

The dominant pattern in 2025-2026 is using a language model to evaluate language model outputs. The practical guidance from Galileo, Arize, Braintrust, and Ragas is consistent:

- Use a *different* and ideally *stronger* model than the one generating content (to avoid self-enhancement bias)
- Give the judge model a precise rubric, not a vague instruction
- Always validate the judge against human labels before trusting it

**The G-Eval Pattern**

DeepEval popularized the **G-Eval** pattern (Liu et al., *"G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment"*, 2023), which asks the judge to reason step-by-step before giving a verdict. Chain-of-thought reasoning significantly improves judge accuracy.

"Rate this on a scale of 1-5" produces worse evaluators than "First explain your reasoning step by step, then give a verdict." Galileo and Braintrust both incorporate this pattern.

**The golden rule: Every evaluator needs human validation**

Use **Cohen's kappa** to measure evaluator-human agreement. A kappa of 0.70 is the minimum for use. Arize AI recommends ≥0.80 before deploying to live traffic — a poorly calibrated evaluator gives false confidence, which is worse than no evaluator.

**Evaluator drift:** W&B Weave recommends monthly recalibration of all LLM-based evaluators. Galileo's alert pattern: if evaluator scores shift more than 5% in a 30-day window without product changes, investigate the evaluator before the product.

## Phase 3: Improve — Fix the Right Thing 🟡

When evaluations reveal a failure, ask first: **which stage of the pipeline introduced this failure?**

| If the failure originates here | Fix this |
|---|---|
| Ingestion / document reading | Improve the PDF parser, OCR, or ASR model |
| Normalization / chunking | Fix how content is split into pieces |
| Retrieval / finding content | Improve search and ranking logic |
| Generation / AI reasoning | Improve the prompt or use a better model |
| Synthesis / audio or video | Improve TTS or video composition |
| Cross-modal mismatch | Improve coordination between output branches |

Galileo's analyses consistently show the most common mistake is routing ingestion and retrieval failures to prompt engineering. The fix improves outputs slightly — but the root cause resurfaces with slightly different inputs.

**The most important insight:** Prompt engineering is the last thing to try, not the first.

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# Part 8: Key Concepts You Need to Know 🟡

## Ingestion — The Most Overlooked Stage

"Ingestion" means reading and extracting content from uploaded files. When it fails, everything downstream measures the wrong system.

Common ingestion failures:

- **OCR errors:** Text-recognition misreads words in scanned documents
- **Caption drops:** PDF parser fails to extract figure captions — the AI doesn't know what figures show
- **ASR hallucination:** Speech-to-text invents words during silence — the "Careless Whisper" problem (*Koenecke et al., FAccT 2024*) showing ASR models fill silence with plausible invented text
- **Layout scramble:** Multi-column PDFs read across columns instead of down
- **Modality blindness:** System processes document text but silently ignores charts and figures

**Ingestion benchmarks:**

- **OCRBench v2** — character and word-level OCR accuracy
- **MMDocBench** — multi-task document understanding
- **DocVQA** — specifically for figure and chart question answering

**Leading parsers (2026):** Docling (IBM, open-source, strong on complex layouts), Google DocAI, Azure Form Recognizer, AWS Textract.

## OCR and ASR — The Two Text Extractors

**OCR (Optical Character Recognition)** converts images of text into machine-readable text.

- Quality measured by **CER (Character Error Rate)** — percentage of characters wrong

**ASR (Automatic Speech Recognition)** converts audio speech into text.

- Quality measured by **WER (Word Error Rate)** — percentage of words wrong

Both have much higher error rates on handwriting, noisy recordings, technical jargon, and accented speech than on clean, standard inputs. Measure these for your specific content types.

**Leading ASR models (2026):** Whisper large-v3 (OpenAI, open-source), Deepgram Nova-3, AssemblyAI Universal-2, Gladia.

## Tracing — How You See Inside Your Pipeline

A **trace** is a complete record of one user request through all pipeline stages. Every major evaluation platform provides tracing:

| Platform | Notable strength |
|---|---|
| **LangSmith** | Audio playback in-trace; native LangChain integration |
| **Arize Phoenix** | Embedding visualization for drift; language-agnostic (OpenTelemetry) |
| **Langfuse** | Self-hostable; full data sovereignty |
| **Braintrust** | Offline + online eval in one platform |
| **W&B Weave** | Strongest compliance certs (SOC 2, HIPAA) |
| **Galileo** | Strongest automated failure clustering |
| **Comet Opik** | Best for teams already using Comet for ML tracking |

## Hallucination — The Industry's Central Problem 🟡

**Hallucination** is when an AI produces content that sounds confident but isn't grounded in reality or provided sources.

Galileo publishes the **Hallucination Index** (quarterly) — a benchmark measuring which frontier models hallucinate least. Key finding: hallucination rates vary dramatically by model *and* by task type. The best model for avoiding hallucination on document QA is often not the best on conversational tasks.

**Two types — requiring different fixes:**

**Intrinsic hallucination (faithfulness failure):** Output contradicts or goes beyond provided source material. → Fix: Better RAG — ensure right content is retrieved and the AI stays grounded in it.

**Extrinsic hallucination (factuality failure):** Output contradicts real-world knowledge. → Fix: Better model selection and fact-checking layers (e.g., Patronus Lynx).

### Embedding Drift 🔴

Arize AI built their monitoring philosophy around this concept. Every piece of content — a document, image, query — can be represented as a list of numbers capturing its meaning (an "embedding"). Similar content clusters together in mathematical space.

When the distribution of user inputs shifts over time, the embedding cluster shifts too. Arize's platform detects this automatically — an early warning that the inputs you're receiving are systematically different from what your system was designed for.

### Consumption Signals — Your Most Honest Metric 🟢

User behavior is your best evaluation signal — and most teams underuse it.

- **Audio:** Listen-through percentage — what % of audio did users listen to?
- **Visual:** Export rate — what % of users exported or copied the diagram?
- **Video:** View-through percentage — how much of the video was watched?

Galileo calls these "downstream signals" and recommends correlating them with technical metrics. If faithfulness is high but listen-through is dropping, either the evaluator has drifted, or there's a quality dimension (pacing, naturalness) that users care about but your stack doesn't capture.

### MOS — How You Measure Audio Quality 🟡

**MOS (Mean Opinion Score)** is a 1-5 scale for audio quality. Below 3.5 feels unpleasant to listen to; 5 is professional-studio quality.

Measure MOS both automatically (UTMOS2, SpeechMOS) and with real humans. Automated MOS needs periodic human recalibration — it drifts across voices and accents.

### Novelty Decay — A Common Trap 🟢

When you launch a new audio or video feature, engagement is often high at first. Then 3-4 weeks later, it drops.

This is **novelty decay** — users tried it because it was new, but the sustained value wasn't there. Braintrust recommends measuring cohort engagement at Day 7 and Day 30. Stable or growing Day 30 engagement = genuine value. Strong Day 7 + weak Day 30 = novelty, not product-market fit.

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## Part 9: The Hallucination Detection Toolkit 🟡

### Decompose-Then-Verify

Used by Galileo, Ragas, and DeepEval:

1. **Decompose** the AI's output into individual atomic claims ("Revenue grew 12% in Q3")
2. **For each claim**, check if it is supported by the source material
3. **Score** as the fraction of claims that are supported

This beats asking a model to rate the whole output at once — a holistic "is this faithful?" question misses subtle individual errors.

### The NLI Approach

**Natural Language Inference (NLI)**: Given a claim and a source sentence, does the source *entail* the claim, *contradict* it, or is it *neutral*? Ragas uses NLI models as a component of their faithfulness metric. Fast and cheap for large-scale checking.

### Hallucination Benchmarks

| Benchmark | What it tests |
|---|---|
| **MMHal-Bench** | 8 categories of multimodal hallucination |
| **POPE** | Object hallucination in visual content |
| **HaloQuest** | Visual hallucination in complex scenes |
| **MMIU** | Multi-image understanding hallucination |
| **FaithDial** | Faithfulness in conversational AI |

These are research benchmarks, not production tools — but running your system on them gives you a calibrated sense of where you sit relative to the field.

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## Part 10: Seven Common Misconceptions 🟢

*These are the wrong beliefs that consistently lead to wasted months. They're listed here explicitly so you can avoid them.*

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**Misconception 1: "I just need to improve my prompt."**

The instinct to reach for prompt engineering first is almost universal — and wrong in most multimodal failure cases. If the failure originated in ingestion (Stage 1), no prompt instruction changes what happened before the AI saw the data. Before touching the prompt, identify which pipeline stage introduced the failure. Prompt engineering is often the right fix — but only after ruling out ingestion, retrieval, and chunking problems first.

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**Misconception 2: "If my automated metrics look good, my product is good."**

Automated evaluators can drift, be miscalibrated, or miss quality dimensions that users actually care about. W&B Weave calls the pattern "judge-user divergence" — when evaluators say green and users are unhappy. Always treat user behavior signals as your final arbiters. If automated metrics and user behavior disagree, believe the users and investigate the evaluators.

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**Misconception 3: "My team's internal testing tells me how real users will experience the product."**

Your team uploads clean, well-formatted files. Real users upload scanned documents with coffee stains, photos of notes taken in dim lighting, and recordings with background noise. Braintrust describes this as a distribution shift between your golden dataset and real traffic. It's the comprehension gulf — and it's nearly universal. The solution is examining real user sessions before writing test cases, not after.

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**Misconception 4: "If the audio and the diagram agree, the outputs are correct."**

Cross-modal consistency means the outputs agree with *each other* — not that they're correct. Consistency is necessary but not sufficient. You must also verify faithfulness against the *original source*, not just between outputs.

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**Misconception 5: "I need sophisticated enterprise tools before I can start evaluating."**

The most valuable evaluation activity — manual trace review — requires zero specialized tooling. Pull 10 real examples. Follow each one through every pipeline stage. Write free-text notes. That's it. The most actionable insights often come from 10 traces and a spreadsheet. Tools amplify discipline; they don't replace it.

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**Misconception 6: "Hallucination is a model problem I can't control."**

You have more control than you think. Intrinsic hallucination (faithfulness failures) is substantially addressed through better RAG — ensuring the right content is retrieved and the AI stays grounded in it. ASR hallucination is addressed through confidence thresholding and silence detection. Caption-drop hallucinations are addressed through better PDF parsers. Most hallucinations that reach users are preventable through pipeline improvements, not just model upgrades.

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**Misconception 7: "A single accuracy score is good enough for reporting."**

A single "accuracy" score collapses faithfulness, factuality, cross-modal consistency, and output quality into one number — and hides failures in each. Ragas demonstrated this empirically: teams using a single score consistently missed faithfulness failures while optimizing for fluency. Always report at least one score per contract, and always pair technical scores with at least one consumption signal.

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# Part 11: The Evaluation Tool Ecosystem — When to Reach for What 🟡

*These recommendations are written as scenarios, not feature lists. The question is: when would you actually reach for each tool?*

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## Scenario A: "We just had a production incident. Users report wrong information, but nothing in our codebase changed."

Reach for **Arize Phoenix** (open-source) or **Arize AI** (enterprise). Their embedding drift visualization will show whether the distribution of user inputs shifted — which is often the actual cause of this pattern. A new type of document your users started uploading may have exposed a gap in your pipeline that your evaluators weren't designed to catch.

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## Scenario B: "We're pre-launch and want to run a systematic quality check on 200 examples before showing anyone."

Reach for **Braintrust** or **DeepEval**. Braintrust is the strongest for dataset versioning and offline evaluation workflow — you can build a scored dataset, iterate on your prompt and pipeline, and

track improvement over time with a full audit trail. DeepEval is a good open-source alternative with 14+ pre-built metrics and a quick setup.

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### Scenario C: "We want to evaluate our RAG pipeline specifically — are we retrieving the right content?"

Reach for **Ragas** (open-source). It was built specifically for RAG evaluation and provides the full RAG Triad out of the box: context relevance, faithfulness, and answer relevance. Setup takes an afternoon.

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### Scenario D: "We're in healthcare (or finance, or law) and need to verify factual claims against real-world knowledge, not just sources."

Reach for **Patronus Lynx**. It was specifically built for factuality evaluation — cross-referencing claims against world knowledge. Pair it with a faithfulness evaluator (Ragas or Galileo) since you need both: claims that are grounded in your sources *and* accurate against the world.

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### Scenario E: "We need to comply with HIPAA, SOC 2, or similar. Our data cannot leave our infrastructure."

Reach for **Langfuse** (self-hosted) or **Weights&Biases Weave** (strongest enterprise compliance certs). Langfuse gives you full observability and evaluation capabilities with complete data sovereignty. W&B Weave has the broadest compliance certifications of any evaluation platform.

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### Scenario F: "We want to understand *why* specific traces are failing — automatically clustered by failure type."

Reach for **Galileo**. Their automated failure clustering groups similar failures together so you can review 50 traces in the time it takes to review 5. For teams past the "10 traces in a spreadsheet" stage, this is the most time-efficient path to root cause analysis.

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### Scenario G: "We're already using LangChain and want tracing with minimal setup."

Reach for **LangSmith**. It integrates natively with LangChain and LangGraph with minimal instrumentation code. For multimodal products, it now supports audio playback directly in the trace view — you can listen to a generated audio output and see its pipeline trace side by side.

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### Scenario H: "We're an ML team already using Comet for experiment tracking."

Reach for **Comet Opik** for LLM evaluation. It extends naturally from your existing Comet workflow, maintaining lineage between your ML experiments and your LLM evaluation runs. Good for teams that want a single platform spanning traditional ML and LLM evaluation.

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**Practical note on tool selection:** Pick one tracing tool and one evaluation tool and use them completely. Two partially-implemented platforms produce worse results than one fully-instrumented one. Arize and Galileo both make this point explicitly in their onboarding documentation — they've seen too many teams spread thin across multiple observability tools with none providing real signal.

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## Part 12: The Six Most Important Rules 🟢

If you remember nothing else from this guide, remember these six rules.

**Rule 1: "It sounds great" is not an evaluation.** Presentation quality (audio naturalness, visual polish) is completely separate from accuracy and faithfulness. Always evaluate them separately.

**Rule 2: Evaluate the pipeline, not just the output.** Following a failure back to its origin stage is more valuable than counting wrong outputs. One ingestion fix can eliminate many downstream failures simultaneously. Braintrust calls this "left-shifting" — moving quality checks earlier where they're cheaper to catch.

**Rule 3: Suspect ingestion first.** One-third of multimodal failures originate in ingestion — before the AI model was involved. Check ingestion before blaming the model.

**Rule 4: One quality score hides too much.** Report separate scores for faithfulness, factuality, cross-modal consistency, and output quality per modality. Ragas demonstrated empirically that a single score consistently hides faithfulness failures while optimizing for fluency.

**Rule 5: Trust human behavior over automated metrics.** If evaluators say great but users stop at minute 3, believe the users. W&B Weave calls this "judge-user divergence." When they disagree, investigate the evaluator.

**Rule 6: Evaluators need maintenance.** The AI models powering your evaluators get updated by vendors. Every major platform recommends monthly recalibration. Check once a month that your evaluators still agree with humans at the same rate they did when you built them.

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# Part 13: Where to Start 🟢

## If You Have a Live Product

### 1: Understand what's failing

*1-1:* Pull 10 real traces. For each, follow the full journey: What did the user upload? What did ingestion extract? What did retrieval find? What did the AI generate? What was the final output? Write free-text notes at each step. Don't categorize yet — just observe. Do this with two people in the room. Disagreements between reviewers are valuable signal about specification gaps.

*1-2:* Review your notes. What failures appear repeatedly? Give each pattern a name (fewer than 15 to start). You're building your first taxonomy.

*1-3:* For each failure pattern, identify which pipeline stage introduced it. This tells you where to focus first.

### 2: Start measuring

*2-1:* Write your quality contracts for your most important output modality. Apply Braintrust's "two rater test": if two independent people rating the same output would disagree 20%+ of the time, your rubric needs refinement before you automate.

*2-2:* Build one binary evaluator for your most common failure pattern. Test it on 20 examples. Calculate agreement with your own human judgment. If below 70%, rewrite the rubric. DeepEval and Ragas both have quick-start guides that can get a basic faithfulness evaluator running in an afternoon.

*2-3:* Start tracking one consumption signal. Listen-through %, export rate, or view-through %. Add it to your dashboard and watch it weekly.

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## If You Don't Have a Product Yet (Pre-Launch Path)

You can do meaningful evaluation work before launch. Here's how.

**Create synthetic traces:** Have 3-5 team members use your product with realistic, messy inputs — the kinds of files real users would submit: low-resolution scans, handwritten notes, recordings with background noise, files exported from legacy software. Treat each session as a real trace.

**Build your evaluation brief first:** Before any development, write down what each of the four contracts means for your specific product. Be specific enough that two people would agree on any given output. This document is the single highest-ROI artifact you can produce pre-launch.

**Red-team your ingestion layer:** Take 20 sample inputs that represent the range of files you expect users to upload. Run them through your ingestion stage and manually compare the extracted output against the originals. This pre-launch ingestion audit has consistently caught the most expensive bugs at the cheapest possible moment.

**Start with DeepEval or Ragas offline:** Both have free tiers and work against a static dataset. Build a baseline evaluation before you have production traffic, so you know what "good" looks like and can detect regressions on Day 1.

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## Closing: The Mental Model That Makes Everything Else Make Sense 🟢

Here is the single mental model that ties everything in this guide together:

**A multimodal AI product is a chain. Its strength is determined by its weakest link — and that weakest link is almost always invisible in the final output.**

Evaluation is the practice of making those weak links visible, one by one, and strengthening them in order of impact.

The final output can look beautiful, sound natural, and feel polished — while carrying failures that originated three or four stages upstream.

The only way to find and fix those failures is to look at the whole chain, not just the finished product.

One team with rigorous evaluation discipline using simple tools will consistently outperform a team with poor discipline using the most sophisticated platform. The discipline matters more than the tool.

Start small. Pull 10 real traces. Follow the chain. Find where things first go wrong. Fix that. Then repeat.

That is multimodal AI product evaluation, at its core.

## Quick Reference: Terms Every Practitioner Needs to Know

| Term | Plain English meaning |
|---|---|
| **Modality** | A type of media: text, image, audio, video, document |
| **Pipeline** | The sequence of stages an AI product runs through |
| **Trace** | A complete record of one user request through all pipeline stages |
| **Span** | One stage within a trace (Arize/LangSmith terminology) |
| **Ingestion** | Reading and extracting content from uploaded files |
| **OCR** | Software that reads text from images — accuracy measured by CER (Character Error Rate) |
| **ASR** | Software that transcribes speech to text — accuracy measured by WER (Word Error Rate) |
| **RAG** | Retrieval-Augmented Generation — AI that retrieves content before answering |
| **RAG Triad** | Three checks for RAG: context relevance, groundedness, answer relevance |
| **Hallucination** | AI producing content that sounds confident but isn't grounded in reality |
| **Intrinsic hallucination** | Output contradicts or exceeds the provided source material |
| **Extrinsic hallucination** | Output contradicts real-world facts |
| **Faithfulness** | Does the output stay within what the sources say? |
| **Factuality** | Does the output accurately describe the real world? |
| **Cross-modal consistency** | Whether different output modalities (audio, visual) agree with each other |
| **Evaluator / Judge / Scorer** | Automated tool that checks output quality |

| | |
|---|---|
| **LLM-as-Judge** | Using an AI model to evaluate another AI model's output |
| **G-Eval** | Evaluation pattern: ask the judge to reason step-by-step before scoring |
| **Cohen's Kappa** | Statistical measure of how well an evaluator agrees with humans (target ≥0.70) |
| **MOS** | Mean Opinion Score — 1-5 scale for audio quality |
| **CER** | Character Error Rate — measure of OCR accuracy |
| **WER** | Word Error Rate — measure of ASR accuracy |
| **Embedding drift** | When the mathematical representation of user inputs shifts over time |
| **Consumption signal** | A user behavior metric: listen-through %, export rate, view-through % |
| **Novelty decay** | Drop in engagement after initial excitement of a new feature wears off |
| **Taxonomy** | Your team's named collection of failure patterns |
| **Offline evaluation** | Testing quality on a fixed dataset before launch |
| **Online evaluation** | Sampling and testing live traffic after launch |
| **Distribution shift** | When the types of inputs users give you change over time |
| **Scoring rubric** | A precise written definition of what "good" means for a specific criterion |
| **Two rater test** | Check: would two independent reviewers agree on this rubric ≥80% of the time? |

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*This guide is an introduction. When you're ready to go deeper, the next booklet - Advanced Field Manual covers each stage, each failure pattern, and each evaluation technique in precise technical detail. The platforms referenced throughout — Galileo, Arize AI, Braintrust, Comet Opik, W&B Weave, Patronus, LangSmith, and Langfuse — all publish extensive documentation that goes further on each individual topic.*

*Key research papers referenced: Liu et al. "G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment" (2023); Koenecke et al. "Careless Whisper: Speech-to-Text Hallucination Harms" (FAccT 2024); Es et al. "RAGAS: Automated Evaluation of Retrieval Augmented Generation" (2023).*